

RW3 Full Modular Run Summary (Combined)

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely $I(1)$ -like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

steps=19

failed=0

modules_with_outputs=19

indexed_sheets=79

tests_tables=9

result_tables=17

Interpretation appears before each module block in the markdown summary.

Table: Run_All_Summary

	step	status	output_dir	error
	sheet_ProducerUA	ok	/Users/getapple/Documents...	
	sheet_ConsumerUA	ok	/Users/getapple/Documents...	
	sheet_EU	ok	/Users/getapple/Documents...	
	sheet_ProZorro	ok	/Users/getapple/Documents...	
	sheet_Silpo	ok	/Users/getapple/Documents...	
	sheet_Novus	ok	/Users/getapple/Documents...	
	sheet_CME	ok	/Users/getapple/Documents...	
model_short_chain_regional		ok	/Users/getapple/Documents...	
model_ardl		ok	/Users/getapple/Documents...	
model_ecm		ok	/Users/getapple/Documents...	
model_nardl		ok	/Users/getapple/Documents...	
model_vecm		ok	/Users/getapple/Documents...	
model_discounts		ok	/Users/getapple/Documents...	
model_intersection_bidire...		ok	/Users/getapple/Documents...	
model_forecast_knn		ok	/Users/getapple/Documents...	
graphs_decomposition		ok	/Users/getapple/Documents...	
graphs_overlay_ln		ok	/Users/getapple/Documents...	
graphs_correlations_lags		ok	/Users/getapple/Documents...	
graphs_brand_region		ok	/Users/getapple/Documents...	

Table: Artifacts_By_Module

module	output_dir	xlsx_count	pdf_count	md_count	png_count	xlsx_files	pdf_files	md_files
graphs_brand_region	/Users/getapple/Documents...	1	1	1	3	graphs_brand_region_outpu...	graphs_brand_region_repor...	graphs_brand_region_repor...
graphs_correlations_lags	/Users/getapple/Documents...	1	1	1	2	graphs_correlations_lags_...	graphs_correlations_lags_...	graphs_correlations_lags_...
graphs_decomposition	/Users/getapple/Documents...	1	1	1	24	graphs_decomposition_outp...	graphs_decomposition_repo...	graphs_decomposition_repo...
graphs_overlay_ln	/Users/getapple/Documents...	1	1	1	21	graphs_overlay_ln_output...	graphs_overlay_ln_report.pdf	graphs_overlay_ln_report.md
model_ardl	/Users/getapple/Documents...	1	1	1	1	model_ardl_output.xlsx	model_ardl_report.pdf	model_ardl_report.md
model_discounts	/Users/getapple/Documents...	1	1	1	2	model_discounts_output.xlsx	model_discounts_report.pdf	model_discounts_report.md
model_ecm	/Users/getapple/Documents...	1	1	1	1	model_ecm_output.xlsx	model_ecm_report.pdf	model_ecm_report.md
model_forecast_knn	/Users/getapple/Documents...	1	1	1	3	model_forecast_knn_output...	model_forecast_knn_report...	model_forecast_knn_report.md
model_intersection_bidire...	/Users/getapple/Documents...	1	1	1	2	model_intersection_bidire...	model_intersection_bidire...	model_intersection_bidire...
model_nardl	/Users/getapple/Documents...	1	1	1	2	model_nardl_output.xlsx	model_nardl_report.pdf	model_nardl_report.md
model_short_chain_regional	/Users/getapple/Documents...	1	1	1	0	primary_chain_consolidate...	primary_chain_consolidate...	primary_chain_consolidate...
model_vecm	/Users/getapple/Documents...	1	1	1	1	model_vecm_output.xlsx	model_vecm_report.pdf	model_vecm_report.md
sheet_CME	/Users/getapple/Documents...	1	1	1	3	sheet_cme_output.xlsx	sheet_cme_report.pdf	sheet_cme_report.md
sheet_ConsumerUA	/Users/getapple/Documents...	1	1	1	3	sheet_consumerua_output.xlsx	sheet_consumerua_report.pdf	sheet_consumerua_report.md
sheet_EU	/Users/getapple/Documents...	1	1	1	3	sheet_eu_output.xlsx	sheet_eu_report.pdf	sheet_eu_report.md
sheet_Novus	/Users/getapple/Documents...	1	1	1	4	sheet_novus_output.xlsx	sheet_novus_report.pdf	sheet_novus_report.md
sheet_ProZorro	/Users/getapple/Documents...	1	1	1	4	sheet_prozorro_output.xlsx	sheet_prozorro_report.pdf	sheet_prozorro_report.md
sheet_ProducerUA	/Users/getapple/Documents...	1	1	1	3	sheet_produceraua_output.xlsx	sheet_produceraua_report.pdf	sheet_produceraua_report.md
sheet_Silpo	/Users/getapple/Documents...	1	1	1	4	sheet_silpo_output.xlsx	sheet_silpo_report.pdf	sheet_silpo_report.md

Table: Sheets_Index

module	xlsx_file	sheet	rows	cols	has_tests_pattern	has_model_results_pattern
graphs_brand_region	graphs_brand_region_outpu...	Brand_Economic_Metrics	1880	10	0	1
graphs_brand_region	graphs_brand_region_outpu...	Brand_IO_Metrics	48	8	0	0
graphs_brand_region	graphs_brand_region_outpu...	Prozorro_ByRegion	171	20	0	0
graphs_correlations_lags	graphs_correlations_lags...	Corr_Matrix	7	8	0	0
graphs_correlations_lags	graphs_correlations_lags...	Correlations	273	10	0	0
graphs_correlations_lags	graphs_correlations_lags...	Lag_Best	39	7	0	0
graphs_correlations_lags	graphs_correlations_lags...	Lag_Profiles	1025	7	0	0
graphs_decomposition	graphs_decomposition_outp...	Decomposition_All	25658	9	0	0
graphs_decomposition	graphs_decomposition_outp...	Decomposition_Index	36	4	0	0
graphs_decomposition	graphs_decomposition_outp...	Decomposition_Summary	36	8	0	0
graphs_overlay_ln	graphs_overlay_ln_output....	BeforeAfterLN_All	25774	8	0	0
graphs_overlay_ln	graphs_overlay_ln_output....	BeforeAfterLN_Index	42	4	0	0
graphs_overlay_ln	graphs_overlay_ln_output....	Overlay_All	13399	10	0	0
graphs_overlay_ln	graphs_overlay_ln_output....	Overlay_Index	12	4	0	0
model_ardl	model_ardl_output.xlsx	ARDL_Summary	0	20	0	1
model_discounts	model_discounts_output.xlsx	Silpo_Discounts_Depth	1	1	0	0
model_discounts	model_discounts_output.xlsx	Silpo_Discounts_Occurrence	1	1	0	0
model_discounts	model_discounts_output.xlsx	Silpo_Transmission_PromoCtrl	1	1	0	0
model_ecm	model_ecm_output.xlsx	ECM_Summary	0	20	0	1
model_forecast_knn	model_forecast_knn_output...	Forecast_Predictions	540	6	0	0
model_forecast_knn	model_forecast_knn_output...	Forecast_Summary	9	7	0	0
model_forecast_knn	model_forecast_knn_output...	Synthetic_Influence_Coeff...	28	7	0	1
model_forecast_knn	model_forecast_knn_output...	Synthetic_Retail_Series	91758	8	0	0
model_forecast_knn	model_forecast_knn_output...	Synthetic_to_Consumer_Link	3	7	0	1
model_forecast_knn	model_forecast_knn_output...	Ultimate_Consumer_Price	145	5	0	0
model_intersection_bidire...	model_intersection_bidire...	Bidirectional_Granger	1	1	0	0
model_intersection_bidire...	model_intersection_bidire...	Bidirectional_Results	1	1	0	0
model_intersection_bidire...	model_intersection_bidire...	CrossTable_Correlations	77	10	0	0
model_intersection_bidire...	model_intersection_bidire...	Intersection_Combination_...	0	0	0	0
model_intersection_bidire...	model_intersection_bidire...	Intersection_Combination_...	1	1	0	0
model_nardl	model_nardl_output.xlsx	NARDL_Summary	0	20	0	1
model_short_chain_regional	primary_chain_consolidate...	Consolidated_Eligibility	48	9	0	0
model_short_chain_regional	primary_chain_consolidate...	Consolidated_ModelCoeffic...	48	20	0	1
model_short_chain_regional	primary_chain_consolidate...	Consolidated_PreTests	144	12	1	1
model_short_chain_regional	primary_chain_consolidate...	Consolidated_ResidualDiag...	48	11	1	1
model_short_chain_regional	primary_chain_consolidate...	Rule_Documentation	1	6	0	0
model_vecm	model_vecm_output.xlsx	VECM_Summary	0	20	0	1
sheet_CME	sheet_cme_output.xlsx	clean	1023	9	0	0
sheet_CME	sheet_cme_output.xlsx	daily_variants	1486	12	0	0
sheet_CME	sheet_cme_output.xlsx	descriptive_stats	3	21	0	0

Table: Tests_Interpretation

module		xlsx_file	sheet	n_rows	i1_like_share	stationary_share	autocorr_risk_share	heterosk_risk_share	non_normal_share	stability_risk_share	interpretation
model_short_chain_regional	primary_chain_consolidate...	Consolidated_PreTests		144	0.0	0.0	0.0	0.0	0.0	0.0	I(1)-like share=0.00, sta...
model_short_chain_regional	primary_chain_consolidate...	Consolidated_ResidualDiag...		48	0.0	0.0	0.0	0.0	0.0	0.0	I(1)-like share=0.00, sta...
	sheet_CME	sheet_cme_output.xlsx	tests	3	0.3333333333333333	0.0	0.3333333333333333	0.0	0.3333333333333333	0.3333333333333333	I(1)-like share=0.33, sta...
sheet_ConsumerUA	sheet_consumerua_output.xlsx		tests	9	1.0	0.0	1.0	1.0	1.0	1.0	I(1)-like share=1.00, sta...
	sheet_EU	sheet_eu_output.xlsx	tests	18	0.3333333333333333	0.0	0.2222222222222222	0.0555555555555555	0.3333333333333333	0.3333333333333333	I(1)-like share=0.33, sta...
	sheet_Novus	sheet_novus_output.xlsx	tests	36	0.0	0.0	0.0	0.0	0.0	0.0	I(1)-like share=0.00, sta...
	sheet_ProZorro	sheet_prozorro_output.xlsx	tests	21	0.19047619047619047	0.09523809523809523	0.09523809523809523	0.0	0.047619047619047616	0.2857142857142857	I(1)-like share=0.19, sta...
sheet_ProducerUA	sheet_producerua_output.xlsx		tests	18	1.0	0.0	1.0	1.0	1.0	1.0	I(1)-like share=1.00, sta...
	sheet_Silpo	sheet_silpo_output.xlsx	tests	36	0.0	0.0	0.0	0.0	0.0	0.0	I(1)-like share=0.00, sta...

Table: Results_Interpretation

module	xlsx_file	sheet	n_rows	mean_abs_coef	coef_positive_share	significance_share_p_lt_005	cointegration_rank_mean	interpretation
graphs_brand_region	graphs_brand_region_output.xlsx	Brand_Economic_Metrics	1880	20.33006410343564	0.4448979591836735	0.1826086956521739	nan	mean coef =20.3301 pos...
model_ardl	model_ardl_output.xlsx	ARDL_Summary	0	nan	nan	nan	nan	Result table has no stand...
model_ecm	model_ecm_output.xlsx	ECM_Summary	0	nan	nan	nan	nan	Result table has no stand...
model_forecast_knn	model_forecast_knn_output.xlsx	Synthetic_Influence_Coeff...	28	0.5	1.0	nan	nan	mean coef =0.5000 posi...
model_forecast_knn	model_forecast_knn_output.xlsx	Synthetic_to_Consumer_Link	3	0.9906802551231674	0.3333333333333333	0.0	nan	mean coef =0.9907 posi...
model_nardl	model_nardl_output.xlsx	NARDL_Summary	0	nan	nan	nan	nan	Result table has no stand...
model_short_chain_regional	primary_chain Consolidate...	Consolidated_ModelCoeffic...	48	nan	nan	nan	nan	Result table has no stand...
model_short_chain_regional	primary_chain Consolidate...	Consolidated_PreTests	144	nan	nan	nan	nan	Result table has no stand...
model_short_chain_regional	primary_chain Consolidate...	Consolidated_ResidualDiag...	48	nan	nan	nan	nan	Result table has no stand...
model_vecm	model_vecm_output.xlsx	VECM_Summary	0	nan	nan	nan	nan	Result table has no stand...
sheet_CME	sheet_cme_output.xlsx	tests	3	nan	nan	0.5	nan	significance share (p<0.0...
sheet_ConsumerUA	sheet_consumerua_output.xlsx	tests	9	nan	nan	0.6666666666666666	nan	significance share (p<0.0...
sheet_EU	sheet_eu_output.xlsx	tests	18	nan	nan	0.5	nan	significance share (p<0.0...
sheet_Novus	sheet_novus_output.xlsx	tests	36	nan	nan	nan	nan	Result table has no stand...
sheet_ProZorro	sheet_prozorro_output.xlsx	tests	21	nan	nan	0.21428571428571427	nan	significance share (p<0.0...
sheet_ProducerUA	sheet_producerua_output.xlsx	tests	18	nan	nan	0.6666666666666666	nan	significance share (p<0.0...
sheet_Silpo	sheet_silpo_output.xlsx	tests	36	nan	nan	nan	nan	Result table has no stand...

Table: Module_Block_Interpretation

module	output_dir	xlsx_files	pdf_files	md_files	png_count	interpretation
graphs_brand_region	/Users/getapple/Documents...	graphs_brand_region_outpu...	graphs_brand_region_repor...	graphs_brand_region_repor...	3	mean coef =20.3301 pos...
graphs_correlations_lags	/Users/getapple/Documents...	graphs_correlations_lags_...	graphs_correlations_lags_...	graphs_correlations_lags_...	2	No explicit test/model ta...
graphs_decomposition	/Users/getapple/Documents...	graphs_decomposition_outp...	graphs_decomposition_repo...	graphs_decomposition_repo...	24	No explicit test/model ta...
graphs_overlay_ln	/Users/getapple/Documents...	graphs_overlay_ln_output...	graphs_overlay_ln_report.pdf	graphs_overlay_ln_report.md	21	No explicit test/model ta...
model_ardl	/Users/getapple/Documents...	model_ardl_output.xlsx	model_ardl_report.pdf	model_ardl_report.md	1	Result table has no stand...
model_discounts	/Users/getapple/Documents...	model_discounts_output.xlsx	model_discounts_report.pdf	model_discounts_report.md	2	No explicit test/model ta...
model_ecm	/Users/getapple/Documents...	model_ecm_output.xlsx	model_ecm_report.pdf	model_ecm_report.md	1	Result table has no stand...
model_forecast_knn	/Users/getapple/Documents...	model_forecast_knn_output...	model_forecast_knn_report...	model_forecast_knn_report.md	3	mean coef =0.5000 posi...
model_intersection_bidire...	/Users/getapple/Documents...	model_intersection_bidire...	model_intersection_bidire...	model_intersection_bidire...	2	No explicit test/model ta...
model_nardl	/Users/getapple/Documents...	model_nardl_output.xlsx	model_nardl_report.pdf	model_nardl_report.md	2	Result table has no stand...
model_short_chain_regional	/Users/getapple/Documents...	primary_chain Consolidate...	primary_chain Consolidate...	primary_chain Consolidate...	0	I(1)-like share=0.00, sta...
model_vecm	/Users/getapple/Documents...	model_vecm_output.xlsx	model_vecm_report.pdf	model_vecm_report.md	1	Result table has no stand...
sheet_CME	/Users/getapple/Documents...	sheet_cme_output.xlsx	sheet_cme_report.pdf	sheet_cme_report.md	3	I(1)-like share=0.33, sta...
sheet_ConsumerUA	/Users/getapple/Documents...	sheet_consumerua_output.xlsx	sheet_consumerua_report.pdf	sheet_consumerua_report.md	3	I(1)-like share=1.00, sta...
sheet_EU	/Users/getapple/Documents...	sheet_eu_output.xlsx	sheet_eu_report.pdf	sheet_eu_report.md	3	I(1)-like share=0.33, sta...
sheet_Novus	/Users/getapple/Documents...	sheet_novus_output.xlsx	sheet_novus_report.pdf	sheet_novus_report.md	4	I(1)-like share=0.00, sta...
sheet_ProZorro	/Users/getapple/Documents...	sheet_prozorro_output.xlsx	sheet_prozorro_report.pdf	sheet_prozorro_report.md	4	I(1)-like share=0.19, sta...
sheet_ProducerUA	/Users/getapple/Documents...	sheet_producerua_output.xlsx	sheet_producerua_report.pdf	sheet_producerua_report.md	3	I(1)-like share=1.00, sta...
sheet_Silpo	/Users/getapple/Documents...	sheet_silpo_output.xlsx	sheet_silpo_report.pdf	sheet_silpo_report.md	4	I(1)-like share=0.00, sta...